

Sub.: _____

SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

44 Final (2018)

DATE: / /



2. (a) Assume that Agrani Bank has a deposit of BDT 300 million and capital of BDT 20 million. Required reserve is 20 percent. The prevailing call money rate and discount rate are 5 percent and 7 percent respectively. The bank disburses BDT 250 million to Ananta group to start its new spinning unit. Agrani bank also buys government treasury bill of BDT 10 million. How much excess reserve does Agrani bank have? Show the balance sheet of Agrani bank in a T-account.

Ans: A T-account is a balance sheet that represents the expansion of deposits by tracking assets owned by the bank and liabilities owed by the bank. Since balance sheet must balance, so too, must T-accounts. Assets and liabilities must balance out, meaning both sides of the T-account should equal the same amount.

Here, The deposits of 300 million and capital of 20 million are both liabilities to the bank.

Sub.: _____

 SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /

P

Loan of 250 million and 10 million treasury bills are asset for the bank.

Required reserve is set against deposits.

Required reserve is 20 percent here. So Agrani bank must keep $-(300 \times 20\%) = 60$ million as reserve.

The balance sheet of Agrani bank in a T-account is shown below: (All measured in BDT)

Agrani Bank

Assets		Liabilities	
Reserves	60 million	Deposits	300 million
Loans	250 million	Bank capital	20 million
Securities	10 million		

Both sides of the T-account are equal.

Agrani bank has no excess reserve. Since the sources of funds (liabilities) are all used up (Assets), there are no excess reserve remaining.

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(b) Prescribe the best way for the bank to meet a deposit outflow of BDT 20 million among all the alternatives. Explain.

Ans: Deposit outflows occur when its depositors withdraw cash from checking or savings accounts or write checks that are deposited in other banks. As we've seen in part (a) that Agrani bank doesn't have any excess reserve, it would have to pay this 20 million of deposit outflow from its reserve. Now, the bank's T-account:

Assets		Liabilities	
Reserves	40 million	Deposits	280 million
Loans	250 million	Bank capital	20 million
Securities	10 million		

Sub.: _____

SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /

P

After 20 million BDT has been withdrawn from deposits and hence reserves, the bank has a problem. It has a reserve requirement of 20% of 280 million or 56 million BDT but it has reserve of 40 million.

To eliminate this shortfall, the bank has 4 basic options:

- (i) Borrowing from other banks or corporation.
- (ii) Selling Securities
- (iii) Borrowing from Bangladesh Bank.
- (iv) Reducing loan amount.

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Borrowing from other banks or corporation, Agrani bank can meet up the shortfall of 16 million reserve. The cost of these borrowings is the interest rate or call money rate.

Agrani bank can also sell off its securities to cover some of its required reserve requirements. As Agrani bank has only 10 million

Sub: _____

SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /



of treasury bills, it would then have to seek from other sources too for the remaining funds.

The bank will incur transaction costs when it sells these securities. As Agrani bank holds Government treasury bills, the transaction cost will be modest.

Borrowing from central bank of Bangladesh, Bangladesh Bank is not valid for Agrani bank.

As Bangladesh bank gives loan only to "Krishi Bank".

Agrani bank can acquire 16 million of reserve from by reducing its loans by this amount. This process is the costliest among four. It may lose customers too by reducing loan.

Sub: _____

SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /

P

So, agrani bank can acquire 16 million reserve either by borrowing from other banks or corporation or ~~from~~ by selling securities. Since Borrowing from other banks incur some costs too, the best way to acquire this fund is to sell its securities of 10 million and borrow \$⁶ million from other banks or corporation at call money rate.

Sub: _____

 SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /

P

(c) How does the outflow affect required reserve of the bank? Show the ^{new} balance sheet of Agrani bank.

Ans: After the deposit outflow of BDT 20 million, the bank's required reserve is now 20% of 280 million or 56 million BDT. Agrani bank did not have any excess reserve. So it paid from its reserve for this outflow.

Assets		Liabilities	
Reserves	40 million	Deposits	280 million
Loans	250 millions	Bank Capital	20 million
Securities	10 million		

~~Now the reserve requirement for Agrani Bank is 20% of 280 million~~

Sub.: _____

SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /



The shortfall in the reserves can be met through selling securities of 10 million BDT and borrowing from other bank or ~~corp~~ corporation (6 million BDT).

The new balance sheet of Agrani bank:

Assets		Liabilities	
Reserves	56 million	Deposits	280 million
Loans	250 million	Bank capital	20 million
		Borrowings from other banks or corporation	6 million

Sub.: _____

SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

44 Final (2018)

DATE: / /



3.(a) Mr. Kabir has withdrawn BDT 1200 from X and keeps up it in his piggy bank forever. Bank X is in a banking system where all banks are loaned up facing 15 percent required rate. Now explain the impact of Mr. Kabir's action on overall credits and deposits in the banking system showing the transactions in first three banks (X, Y, and Z). What is the total creation/destruction of money in the banking system?

Ans: Suppose, initially bank X had deposits of BDT 1200. ~~but~~ The reserve ^{requirement} ~~required~~ is 15%.

All banks in the banking system are loaned up - meaning banks have loaned out all excess reserve available to them.

Sub.: _____

 SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /



Bank X

Assets		Liabilities	
Reserves	180	Deposits	1200
Loans	1020		

Mr Kabir's withdrawal of BDT 1200 from bank X has the following effects:

Bank X held only 180 BDT as reserve. So upon withdrawal, it will have a net deficiency of BDT 1020.

Suppose, Bank X sells securities in the amount of BDT 1020 to raise the necessary funds.

Bank X

Assets		Liabilities	
Reserves	-180	Deposits	-1200
Securities	-1020		
Cash	+1020		

Sub: _____

 SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /

P

Assuming customers of bank Y buy these securities from bank X by withdrawing BDT 1020 from bank Y. Since bank Y was also loaned up and held 15% of its reserve - it will have a net deficiency of 867 BDT (15% of 1020 = 153 BDT is already in reserve of bank Y. So net deficiency $1020 - 153 = 867$ BDT).

To meet the deficiency, bank Y will also sell securities in the amount of net deficiency (867 BDT).

Bank Y

Assets		Liabilities	
Reserves	-153	Deposits	-1020
Securities	-867		
Cash	#		

Sub: _____

SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /

P

Customers of Bank Z buy securities from bank Y by withdrawing 867 BDT.

Bank Z will also have net deficiency. It held 15% of 867 = 130.05 BDT as reserve.

So net deficiency $867 - 130.05 = 736.95$ BDT.

To raise funds, bank Z will also sell securities in the amount of 736.95 BDT.

Bank Z

Assets		Liabilities	
Reserves	- 130.05	Deposits	- 867
Securities	- 736.95		

Total Destruction of money in the banking

$$\text{system} = \frac{-\text{Initial Withdrawal}}{\text{reserve requirement}}$$

$$= \frac{-1200}{.15}$$

$$= -8000$$

8000 BDT is destroyed from the banking system.

Sub.: _____

SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /

P

(b) Financial statements of Bank X show that its profit in 2017 was BDT 250 million and it paid BDT 50 million as corporate tax. The statement also revealed that assets of Bank stood at BDT 2000 million and equity capital was 400 million in 2017. Calculate bank X's return on assets (ROA), return on equity (ROE) and equity multiplier (EM) for 2017.

Ans: Here, Bank X's net profit after taxes =
(250 - 50) million BDT
= 200 million BDT

Assets = 2000 million BDT

Equity capital = 400 million BDT

Return on assets (ROA) is a basic measure of bank profitability.

$$ROA = \frac{\text{net profit after taxes}}{\text{assets}}$$

Sub.: _____

SAT	SUN	MON	TUE	WED	THU	FRI
☐	☐	☐	☐	☐	☐	☐

DATE: / /



$$= \frac{200}{2000}$$

$$= 0.1$$

Return on assets (ROA) of bank X is 10%.

Return on equity (ROE) measures how well the owners are doing on their investment.

$$ROE = \frac{\text{net profit after taxes}}{\text{equity capital}}$$

$$= \frac{200}{400}$$

$$= 0.5$$

Return on equity (ROE) of bank X is 50%.

$$\text{Equity multiplier} = \frac{\text{assets}}{\text{equity capital}}$$

$$= \frac{2000}{400}$$

$$= 5$$

Equity multiplier of bank X is 5.

Sub: _____

SAT ☐ SUN ☐ MON ☐ TUE ☐ WED ☐ THU ☐ FRI ☐

DATE: / /



(c) Assume that Bank Y's ROA was same as Bank X in 2017, but the equity multiplier of Bank Y was 8, then what was the ROE of Bank Y in 2017?

Ans:

Bank X's ROA in 2017 = 10%

So Bank Y's ROA in 2017 was also 10%.

Equity multiplier (EM) of Bank Y in 2017 = 8

$\therefore$ ROE of Bank Y = ROA $\times$ EM

$$= 8 \times 10\%$$

$$= 80\%$$

$\therefore$ Return of equity of Bank Y in 2017 was 80%.